ratvalue — Mathematical Reference

Exact formulas as implemented, module by module

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Contents

1 Arithmetic model

Exact rational arithmetic. All monetary values are stored as `ratmoney::Currency` (integer units in the smallest denomination, e.g. cents) with an exact `Rational{num, den}` rate. All dimensionless quantities (rates, ratios, growth rates) are stored as `ratmoney::Rational{num, den}` with gcd reduction over `__int128`.

The only exception: discount factors. Expressions of the form $(1+r)^{-t}$ are inherently irrational; they are computed in `double` and the result is rounded back to integer currency units.

> **Implementation note: double used in every PV sum**
>
> Any line involving `std::pow(1.0 + r, -t)` is `double`. All multipliers applied to `Currency` values before discounting are first converted to `double` via `detail::to_double()`. Terminal value multipliers such as $(1+g)/(\text{rate}-g)$ are computed as exact `Rational` and then applied via `Currency::scale()` *before* the discount step; the final PV discounting is `double`.

Terminal value multiplier (shared pattern). For any Gordon-Growth terminal step $\text{TV} = X_N \cdot \frac{1+g}{r-g}$, the multiplier is constructed as an exact `Rational`:

$$\text{TV\_mul} = \frac{(g.\text{den} + g.\text{num}) \cdot r.\text{den}}{r.\text{num} \cdot g.\text{den} - g.\text{num} \cdot r.\text{den}} \tag{1}$$

where r is WACC, K_u, or K_e depending on the model. This is exact over `__int128` before `Rational` reduction.

2 EBIT normalization

> **Damodaran ref: Investment Valuation, ch. 8 and 18**
>
> Cyclical or non-recurring items distort a single-year EBIT.

Method 1 — Historical average.

$$\overline{\text{EBIT}} = \frac{1}{n}\sum_{i=1}^{n} \text{EBIT}_i \qquad n \geq 2 \text{ [exact]} \tag{2}$$

Implemented as exact integer sum followed by `Currency::scale({1,n})`.

Method 2 — Normalized margin.

$$\text{EBIT} = \text{revenue}_{\text{current}} \times m \qquad m \geq 0 \text{ [exact]} \tag{3}$$

Implemented via `Currency::scale(m)` where m is a `Rational`.

3 CAPM — cost of equity

> **Damodaran ref: Investment Valuation, ch. 4; "Equity Risk Premiums", Damodaran (2023)**

$$K_e = R_f + \beta \cdot \text{ERP}_{\text{mature}} + \lambda \cdot \text{CRP} \tag{4}$$

All four parameters and the result are `Rational`; fully exact. $\lambda = 1$ and $\text{CRP} = 0$ by default (developed market).

For Brazil: $\text{CRP} \approx \text{sovereign spread} \times (\sigma_{\text{equity}}/\sigma_{\text{bond}})$.

Implementation note:

Computed in three exact steps: $\beta \times \mathrm{ERP}_{\mathrm{mature}}$, $\lambda \times \mathrm{CRP}$, $R_f + \beta\mathrm{ERP} + \lambda\mathrm{CRP}$.

4 Beta — Hamada equation and bottom-up beta

Damodaran ref: Investment Valuation, ch. 4; Hamada (1972)

Lever / unlever.

$$\beta_L = \beta_U \cdot [1 + (1-t) \cdot D/E] \text{ [exact]} \tag{5}$$

$$\beta_U = \beta_L \; / \; [1 + (1-t) \cdot D/E] \text{ [exact]} \tag{6}$$

Bottom-up beta.

1. For each comparable i: $\beta_{U,i} = \beta_{L,i}/[1 + (1 - t_{\text{sector}}) \cdot (D/E)_i]$. Computed in `double`.
2. $\bar{\beta}_U = \frac{1}{n}\sum_i \beta_{U,i}$ [double]
3. Re-lever to target structure: $\beta_L = \bar{\beta}_U[1 + (1 - t_{\text{target}}) \cdot (D/E)_{\text{target}}]$ [double]
4. Result approximated to 4 decimal places as `Rational`: $\beta_L \approx \lfloor \beta_L \times 10^4 \rceil / 10^4$.

Implementation note: double in bottom-up

Steps 1–3 are entirely in `double` (market data; 4-decimal precision is sufficient). Only the final rounding back to `Rational` is exact.

5 Synthetic K_d

Damodaran ref: "Ratings, Interest Coverage Ratios and Default Spreads", Damodaran (Jan 2023)

$$K_d = R_f + \text{spread}(\text{rating}(\text{ICR})) \tag{7}$$

$$\text{ICR} = \text{EBIT}/\text{Interest expense} \tag{8}$$

The spread is looked up from a 15-level credit rating table (Moody's/S&P simplified scale). Two tables available: `large_firm` (default) and `small_firm` (stricter ICR thresholds).

Rating	$\text{ICR}_{\min}$ (large)	Spread (bps)	Rating	$\text{ICR}_{\min}$ (small)	Spread (bps)
AAA	8.50	63	AAA	12.50	63
AA	6.50	78	AA	9.50	78
A+	5.50	98	A+	7.50	98
A	4.25	108	A	6.00	108
A−	3.00	122	A−	4.50	122
BBB	2.50	156	BBB	4.00	156
BB+	2.25	200	BB+	3.50	200
BB	2.00	240	BB	3.00	240
B+	1.75	350	B+	2.50	350
B	1.50	375	B	2.00	375
B−	1.25	500	B−	1.50	500
CCC	0.80	600	CCC	1.25	600
CC	0.65	700	CC	0.80	700
C	0.20	800	C	0.50	800
D	$-\infty$	1200	D	$-\infty$	1200

Implementation note:

The spread (in basis points) is stored as `Rational{spread_bps, 10000}`; $K_d = R_f + \text{spread}$ is exact. When Interest expense = 0: ICR = $+\infty \Rightarrow$ AAA.

6 WACC

Damodaran ref: Investment Valuation, ch. 8

$$\text{WACC} = \frac{E}{V}\,K_e + \frac{D}{V}\,K_d\,(1-t) \tag{9}$$

All five inputs (*equity weight*, *debt weight*, K_e, K_d, t) and the result are `Rational`; fully exact.

Implementation note:

$(1-t) = (t.\text{den} - t.\text{num})\,/\,t.\text{den}$ (exact). Two intermediate terms: $D/V \cdot K_d \cdot (1-t)$ and $E/V \cdot K_e$, then summed.

7 FCFF

Damodaran ref: Investment Valuation, ch. 11

$$\text{FCFF} = \text{EBIT}(1-t) + D\&A - \text{CapEx} - \Delta\text{NWC} \tag{10}$$

All values are `Currency`; $(1-t)$ applied via `Currency::scale({t.den-t.num, t.den})`. Fully exact.

Multi-stage projection. For stage k with growth rate g_k and n_k periods, starting from FCFF_0:

$$\text{FCFF}_{t+1} = \text{FCFF}_t \cdot (1+g_k), \quad t = 0,\ldots,n_k - 1 \tag{11}$$

Factor $(1+g_k)$ computed as `Rational{g_k.den + g_k.num, g_k.den}`; `Currency::scale` is exact. Each projected value is stored as an exact `Currency`.

8 DCF (FCFF-based)

Damodaran ref: Investment Valuation, ch. 12

Two-stage model.

$$\mathrm{PV}(\mathrm{FCFF}_t) = \mathrm{FCFF}_t \cdot (1 + \mathrm{WACC})^{-t} \text{ [double]} \tag{12}$$

$$\mathrm{TV} = \mathrm{FCFF}_N \cdot \frac{1+g}{\mathrm{WACC} - g} \text{ [exact] (multiplier, eq. 1)} \tag{13}$$

$$\mathrm{PV}(\mathrm{TV}) = \mathrm{TV} \cdot (1 + \mathrm{WACC})^{-N} \text{ [double]} \tag{14}$$

$$\mathrm{EV} = \sum_{t=1}^{N} \mathrm{PV}(\mathrm{FCFF}_t) + \mathrm{PV}(\mathrm{TV}) \tag{15}$$

$$\mathrm{Equity} = \mathrm{EV} - D_{\mathrm{net}} \tag{16}$$

$$P = \mathrm{Equity}/n_{\mathrm{shares}} \tag{17}$$

Guard: $\mathrm{WACC} > g$ is enforced before computation; returns `WACCLEGrowth` otherwise.

Implementation note:

TV multiplier is exact (eq. 1 with $r = \mathrm{WACC}$). All PV sums, PV(TV) and the final EV are accumulated in `double` and rounded to integer `Currency` units at the end. D_{net} subtraction and per-share division use exact `Currency` and `Rational{1,n_shares}`.

9 FCFE and equity DCF

Damodaran ref: Investment Valuation, ch. 13

$$\mathrm{FCFE} = \mathrm{NI} - (1-\delta)(\mathrm{CapEx} - D\&A + \Delta\mathrm{NWC}) + \Delta D_{\mathrm{net}} \tag{18}$$

where $\delta = D/V$ (debt ratio). Fully exact; $(1-\delta)$ computed as `Rational{delta.den - delta.num, delta.den}`.

Equity DCF. Identical structure to the FCFF DCF (Section 8), with K_e replacing WACC and no net-debt subtraction:

$$\mathrm{PV}(\mathrm{FCFE}_t) = \mathrm{FCFE}_t \cdot (1 + K_e)^{-t} \text{ [double]} \tag{19}$$

$$\mathrm{TV}_{\mathrm{FCFE}} = \mathrm{FCFE}_N \cdot \frac{1+g}{K_e - g} \text{ [exact] (eq. 1 with } r = K_e) \tag{20}$$

$$\mathrm{Equity} = \sum_t \mathrm{PV}(\mathrm{FCFE}_t) + \mathrm{PV}(\mathrm{TV}_{\mathrm{FCFE}}) \tag{21}$$

$$P = \mathrm{Equity}/n_{\mathrm{shares}} \tag{22}$$

10 DDM — multi-stage and H-Model

Damodaran ref: Investment Valuation, ch. 13–14

Multi-stage DDM. Dividends are projected using the same growth engine as FCFF (Section 7, eq. 11).

$$P = \sum_{t=1}^{N} D_t(1+K_e)^{-t} + D_N \cdot \frac{1+g_l}{K_e - g_l} \cdot (1+K_e)^{-N} \tag{23}$$

TV multiplier exact (eq. 1 with $r = K_e$); PV sums in `double`.

H-Model (closed form). Growth declines linearly from g_h to g_l over $2H$ years:

$$P = D_0 \cdot \frac{(1+g_l) + H\,(g_h - g_l)}{K_e - g_l} \tag{24}$$

Implementation note: fully exact

All four sub-expressions — $(1+g_l)$, $H(g_h - g_l)$, numerator, $(K_e - g_l)$ — are computed as exact `Rational`. Final multiplier applied to D_0 via `Currency::scale`; no `double` anywhere. Constraint: $H > 0$ and $K_e > g_l$.

11 Terminal value extensions

Damodaran ref: Investment Valuation, ch. 12; "Consistent Terminal Value"

Consistent terminal value (ROIC-based). Derives the stable-period reinvestment rate from g and ROIC_s, then applies it to NOPAT_s:

$$\mathrm{RR}_s = g_s/\mathrm{ROIC}_s \text{ [exact]} \tag{25}$$

$$\mathrm{TV} = \mathrm{NOPAT}_s \cdot \frac{1 - \mathrm{RR}_s}{\mathrm{WACC} - g_s} \text{ [exact]} \tag{26}$$

Implementation note:

$(1 - \mathrm{RR}_s) = 1 - g_s/\mathrm{ROIC}_s$; $(\mathrm{WACC} - g_s)$; ratio of the two; all exact `Rational`. Applied to `Currency` via `scale()`. Requires $\mathrm{ROIC}_s > 0$ and $\mathrm{WACC} > g_s$.

Consistency check. Returns three diagnostics:

$$\begin{aligned} \text{implied\_RR} &= g/\mathrm{ROIC} \text{ [exact]} \\ \text{spread} &= \mathrm{ROIC} - \mathrm{WACC} \text{ [exact]} \\ \text{value\_creating} &\Leftrightarrow \mathrm{ROIC} > \mathrm{WACC} \\ \text{feasible} &\Leftrightarrow 0 \le g/\mathrm{ROIC} \le 1 \end{aligned}$$

TV by multiple.

$$\mathrm{TV} = \mathrm{EBITDA}_{\mathrm{terminal}} \times \text{multiple [exact]} \tag{27}$$

12 APV — Modigliani-Miller and Miles-Ezzell

Damodaran ref: Investment Valuation, ch. 15; Miles & Ezzell (1980)

Unlevered firm value (shared by both variants).

$$V_U = \sum_{t=1}^{N} \mathrm{FCFF}_t (1+K_u)^{-t} + \mathrm{FCFF}_N \cdot \frac{1+g}{K_u - g} \cdot (1+K_u)^{-N} \tag{28}$$

TV multiplier exact (eq. 1 with $r = K_u$); discount sums in `double`.

MM tax shield (permanent debt).

$$\mathrm{PV(TS)}_{\mathrm{MM}} = t \cdot D \text{ [exact]} \tag{29}$$

$$\mathrm{APV} = V_U + \mathrm{PV(TS)} \tag{30}$$

$$\mathrm{Equity} = \mathrm{APV} - D_{\mathrm{net}} \tag{31}$$

Implementation note: exact Currency::scale

$\mathrm{PV(TS)}_{\mathrm{MM}} = D.\mathrm{scale}(t)$; no `double`.

Miles-Ezzell tax shield (constant D/V, rebalanced debt).

$$\mathrm{PV(TS)}_{\mathrm{ME}} = t \cdot K_d \cdot D \cdot \frac{1+K_u}{(1+K_d)(K_u - g)} \tag{32}$$

Implementation note: exact multiplier

The scalar multiplier $\mu = tK_d(1+K_u) \,/\, [(1+K_d)(K_u - g)]$ is built step-by-step as exact `Rational`: tK_d, $(1+K_u)$, $(1+K_d)$, $(K_u - g)$, then $\mu = (tK_d)(1+K_u) \,/\, [(1+K_d)(K_u - g)]$. Applied to D via `Currency::scale`; no `double`.

13 Excess Returns / EVA

Damodaran ref: Investment Valuation, ch. 31; Stewart (1991)

Single-stage (closed form).

$$V = \mathrm{IC} \cdot \frac{\mathrm{ROIC} - g}{\mathrm{WACC} - g} \text{ [exact]} \tag{33}$$

$$\mathrm{PV(EVA)} = \mathrm{IC} \cdot \frac{\mathrm{ROIC} - \mathrm{WACC}}{\mathrm{WACC} - g} \text{ [exact]} \tag{34}$$

Multi-stage. IC_t is projected using the same growth engine as FCFF (eq. 11).

$$\mathrm{EVA}_t = (\mathrm{ROIC} - \mathrm{WACC}) \cdot \mathrm{IC}_t \text{ [exact]} \tag{35}$$

$$\mathrm{PV}(\mathrm{EVA}_t) = \mathrm{EVA}_t \cdot (1 + \mathrm{WACC})^{-t} \text{ [double]} \tag{36}$$

$$\mathrm{PV}(\mathrm{EVA}_\infty) = \frac{(\mathrm{ROIC}_s - \mathrm{WACC}) \cdot \mathrm{IC}_N}{\mathrm{WACC} - g_s} \cdot (1 + \mathrm{WACC})^{-N} \text{ [double]} \tag{37}$$

$$V = \mathrm{IC}_0 + \sum_t \mathrm{PV}(\mathrm{EVA}_t) + \mathrm{PV}(\mathrm{EVA}_\infty) \tag{38}$$

$(\mathrm{ROIC} - \mathrm{WACC})$ exact `Rational`; IC_t exact `Currency`; EVA_t via `Currency::scale`; discounting in `double`.

14 Optimal capital structure

Damodaran ref: Investment Valuation, ch. 15–16

For $i = 0, 1, \dots, N-1$:

$$d_i = i/N \quad \text{(debt ratio, } \texttt{Rational\{i,N\}}) \tag{39}$$
$$D/E_i = i/(N-i) \quad (\texttt{Rational\{i, N-i\}}, \text{ or } 0 \text{ for } i = 0) \tag{40}$$
$$\beta_{L,i} = \beta_U[1 + (1-t)\, D/E_i] \text{ [exact] (eq. 5)} \tag{41}$$
$$K_{ei} = R_f + \beta_{L,i}\, \text{ERP} + \lambda\, \text{CRP} \text{ [exact] (eq. 4)} \tag{42}$$
$$K_{di} = R_f + \text{spread}(\text{ICR}_i) \text{ [exact]} \tag{43}$$

K_{di} is solved iteratively (circular dependency ICR $\leftrightarrow K_d$):

$$\text{ICR}_i^{(k)} = \frac{\text{EBIT}}{d_i \cdot V \cdot K_{d_i}^{(k)}}, \qquad K_{d_i}^{(k+1)} = R_f + \text{spread}\Big(\text{ICR}_i^{(k)}\Big) \tag{44}$$

Initial $K_d^{(0)} = R_f$; converges in ≤ 10 iterations (tolerance 10^{-9}).

$$\text{WACC}_i = (1-d_i)\, K_{ei} + d_i\, K_{di}\, (1-t) \text{ [exact] (eq. 9)} \tag{45}$$
$$i^* = \arg\min_i \text{WACC}_i \text{ [double] (comparison in } \texttt{double}) \tag{46}$$

Implementation note:

The ICR iteration loop runs in `double`; the converged K_d value is the `Rational` returned by `synthetic_cost_of_debt_from_icr`. WACC at each point is exact `Rational`; the minimum-search comparison uses `double` to avoid `__int128` overflow.

15 Justified multiples

Damodaran ref: Investment Valuation, ch. 17–20

All derived from the Gordon Growth Model.

$$g = \text{ROE} \cdot (1 - \text{payout}) \text{ [exact]} \tag{47}$$
$$P/E = \frac{\text{payout}}{K_e - g} \text{ [exact]} \tag{48}$$
$$P/BV = \text{ROE} \cdot P/E \text{ [exact]} \tag{49}$$
$$\text{RR} = g/\text{ROIC} \text{ [exact]} \tag{50}$$
$$EV/\text{EBITDA} = \frac{(1-t)(1-\text{RR})}{\text{WACC} - g} \text{ [exact]} \tag{51}$$

Guards: $K_e > g$ and WACC $> g$.

16 Fundamental growth

Damodaran ref: Investment Valuation, ch. 11 and 13

$$\mathrm{ROIC} = \mathrm{NOPAT}/\mathrm{IC} \text{ [exact] (via } \texttt{Currency} \text{ units)} \tag{52}$$
$$\mathrm{RR}_{\mathrm{firm}} = (\mathrm{CapEx} - D\&A + \Delta\mathrm{NWC})/\mathrm{NOPAT} \text{ [exact]} \tag{53}$$
$$g_{\mathrm{firm}} = \mathrm{ROIC} \cdot \mathrm{RR}_{\mathrm{firm}} \text{ [exact]} \tag{54}$$
$$\mathrm{ROE} = \mathrm{NI}/\mathrm{BV} \text{ [exact]} \tag{55}$$
$$g_{\mathrm{equity}} = \mathrm{ROE} \cdot (1 - \mathrm{payout}) \text{ [exact]} \tag{56}$$

Implementation note:

ROIC and RR are `Rational` constructed from raw integer units of the `Currency` operands; no `double`.

17 Scenario valuation

Damodaran ref: Investment Valuation, ch. 24; "Valuing Young and Start-up Firms"

For each scenario i with probability P_i, terminal FCFF F_i, growth g_i, cost of capital WACC_i, and horizon T_i:

$$V_i = \frac{F_i(1+g_i)}{\mathrm{WACC}_i - g_i} \cdot (1 + \mathrm{WACC}_i)^{-T_i} \tag{57}$$
$$\mathbb{E}[\mathrm{EV}] = \sum_i P_i \, V_i \text{ [double]} \tag{58}$$
$$\mathrm{EV} = \mathbb{E}[\mathrm{EV}] \cdot P_{\mathrm{surv}} + V_{\mathrm{fail}} \cdot (1 - P_{\mathrm{surv}}) \text{ [double]} \tag{59}$$
$$\mathrm{Equity} = \mathrm{EV} - D_{\mathrm{net}} \tag{60}$$

TV multiplier exact (eq. 1); all V_i and subsequent steps in `double`.
Guard: $\sum_i P_i = 1$ (tolerance 10^{-6}).

18 Equity as option — Merton model

Damodaran ref: Investment Valuation, ch. 30; Merton (1974)

Models equity as a European call option on firm assets with strike $= D$ (face value):

$$d_1 = \frac{\ln(V/D) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \tag{61}$$
$$d_2 = d_1 - \sigma\sqrt{T} \tag{62}$$
$$E = V\,\mathcal{N}(d_1) - D\,e^{-rT}\,\mathcal{N}(d_2) \tag{63}$$
$$P(\mathrm{default}) = \mathcal{N}(-d_2) \tag{64}$$

where $\mathcal{N}(x) = \frac{1}{2}\,\mathrm{erfc}(-x/\sqrt{2})$ (via `std::erfc`).
All computation in `double`; no `Rational` arithmetic.

Implementation note: inputs are assets, not equity

V = asset value, σ = asset volatility. These are not directly observable. Deriving them from observable equity market data $\{E_{\text{mkt}}, \sigma_E\}$ requires solving the system $\{E = \text{BS}(V, \sigma);\ \sigma_E E = \mathcal{N}(d_1)\sigma V\}$ iteratively — this step is *not* implemented; the caller must supply (V, σ) directly.

19 Bank FCFE — regulatory capital approach

Damodaran ref: Investment Valuation, ch. 22–23 (Basel III)

Reinvestment for a bank = increase in regulatory capital required by risk-weighted asset growth (Basel III Tier 1):

$$\text{equity\_RR} = \frac{(\text{RWA}_{\text{proj}} - \text{RWA}_{\text{cur}}) \cdot t_1}{\text{NI}} \text{ [exact]} \tag{65}$$

$$\text{FCFE}_{\text{bank}} = \text{NI} \cdot (1 - \text{equity\_RR}) \text{ [exact]} \tag{66}$$

equity_RR is a `Rational` built from the integer `Currency` units of ΔRWA and NI. Subsequent DCF uses `compute_fcfe_dcf` (Section 9).

20 Arithmetic precision summary

Operation	Arithmetic	Note
CAPM, WACC, Hamada lever/unlever	exact `Rational`	
FCFF, FCFE formulas	exact `Currency`	
FCFF/IC projection $(1+g)^n$	exact per step	accumulated multiplication
TV multiplier $(1+g)/(r-g)$	exact `Rational`	
$t \times D$ (MM tax shield)	exact `Currency::scale`	
ME tax shield multiplier	exact `Rational`	
H-Model price	exact `Rational+Currency`	no `double`
Justified multiples	exact `Rational`	
ROIC, RR, ROE, g	exact `Rational`	from `Currency` units
Bank RR	exact `Rational`	from `Currency` units
EBIT normalization	exact `Currency`	
PV sums $\sum F_t(1+r)^{-t}$	`double`	$(1+r)^{-t}$ irrational
PV(TV) $(1+r)^{-N}$	`double`	irrational
Bottom-up beta (steps 1–3)	`double`	market data
ICR iteration in cap. struct.	`double`	convergence loop
WACC min-search comparison	`double`	avoid `int128` overflow
Merton model (all)	`double`	Black-Scholes
Scenario EV accumulation	`double`	weighted sum

21 Damodaran fidelity checklist

The following items deviate from or extend the textbook treatment and should be verified when comparing model output to Damodaran's worked examples.

1. **FCFF sign convention.** CPV, selling expenses, G&A and other operating items enter as signed values (negative for costs). The pipeline feeds them already signed; verify that all cost fields carry the correct sign before passing to `compute_fcff`.

2. **D&A proxy in the pipeline.** `brapi_fetch.py` approximates D&A as `incomeFromOperations` − `netIncome`; this includes deferred taxes and other non-cash items. For precise debugging, compare against the reported depreciation line from the notes to the financial statements.

3. **CapEx from cash flow statement.** CapEx = |operatingCashFlow − freeCashFlow| (brapi); falls back to |investmentCashFlow| if zero. Both are proxies.

4. **H-Model requires $H > 0$.** The formula degenerates at $H = 0$; the library returns `InvalidInput`.

5. **TV consistency check does not prevent computation.** `check_terminal_consistency` is informational only; `compute_dcf` uses the Gordon formula regardless of ROIC. Use `consistent_terminal_value` explicitly if you want the ROIC-adjusted TV.

6. **Bottom-up beta precision.** Unlevering and re-levering are in `double`; the result is rounded to 4 decimal places. For an exact treatment, supply β_L directly as a `Rational`.

7. **Capital structure: circular K_d.** The iteration uses ≤ 10 steps with tolerance 10^{-9}. For unusual EBIT/FV ratios it may not converge; inspect `interest_coverage` in each `CapitalStructurePoint`.

8. **Merton model: asset value must be supplied.** See Section 18. The library does not solve for (V, σ_V) from $(E_{\text{mkt}}, \sigma_E)$.

9. **Scenario valuation: flat terminal FCFF.** Each scenario describes a single terminal state. There are no intermediate projected years; V_i is purely the discounted TV. For richer scenario paths, chain with FCFF projection first.

10. **WACC-based CDI model (pipeline only).** The Python pipeline does *not* use the standard CAPM implementation above. Instead it uses a CDI-based model: $K_u = \text{CDI} + \pi_E \cdot E/V$, $K_e = K_u + (K_u - K_d(1-t)) \cdot D/E$, $\text{WACC} = E/V \cdot K_e + D/V \cdot K_d(1-t)$. This is defined entirely in `prepare_inputs.py` and is independent of the C++ modules in Sections 3–6.